

① a) $m \rightarrow$ nr de persoane
 $ms \rightarrow$ nr de simulări

def same-birthday-simulations
 (m, ms):

count = 0

for _ in range(ms):

birthdays $\rightarrow$ generează m
 nr aleatoare între 1 și 365

Folosind funcția randint

if len(birthdays) != len(set(birthdays)).

count += 1

return count / ms

c) xs = range(2, 51)

ys = [same-birthday-simulations
 (m, 1000) for m in xs]

plot(xs, ys, 'rx')

show() $\rightarrow$ afișează toate figurile

$$\varphi_m(A) = \frac{n_m(A)}{m}$$

$$\lim_{m \rightarrow \infty} \varphi_m(A) = p(A)$$

$n_m(A)$ = nr. de realizări
 ale ev A în m simulări

b) Ev contrar. $\bar{A}$. Toate
 persoanele au zile de
 naștere diferite

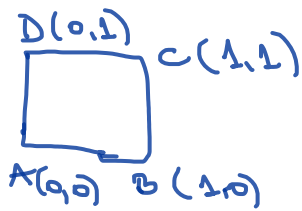
Pt. pers. 1. 365 posibilități pt
 a alege ziua de naștere

Pt. pers. 2 364

Pt. pers m. 365 - m + 1

$$P(\bar{A}) = \frac{365}{365} \cdot \frac{364}{365} \cdot \dots \cdot \frac{365-m+1}{365}$$

$$P(A) = 1 - P(\bar{A})$$



$$l_{\text{patrat}} = 1$$

def puncte_patrat(N, count):
 axis('square')
 axis([0, 1, 0, 1])
 count = 0
 A, B, C, D, E = [0, 0], [1, 0],
 [1, 1], [0, 1], [0.5, 0.5]

match count:

case 1

for _ in range(N):

F = [random(), random()]

if ...

count += 1

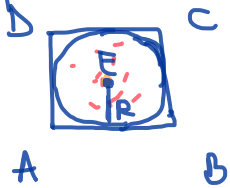
plot(F[0], F[1], 'rx')

else:

plot(F[0], F[1], 'bo')

return count / N

i) puncte în interiorul cercului tangent



$$R_{\text{cerc}} = \frac{1}{2}, E(0.5, 0.5)$$

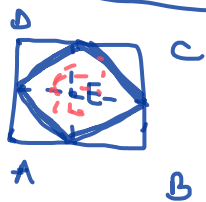
$$\text{dist}(F, E) < R \Rightarrow$$

$$\text{dist}(F, E) < 0.5$$

$$p_{\text{Geometrică}} = \frac{A_{\text{cerc}}}{A_{\text{patrat}}} = \frac{\pi \cdot R^2}{l^2} = \frac{\pi}{4}$$

ii) Punctele mai apropiate de centrul pătratului decât de vîrfurile sale

$$\text{dist}(F, E) < \min([\text{dist}(F, A), \text{dist}(F, B), \text{dist}(F, C), \text{dist}(F, D)])$$



$$p_{\text{Geometrică}} = \frac{A_{\text{"nomi"}}}{A_{\text{patrat}}} = \frac{1}{2}$$

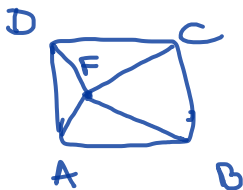
iii) Pct formarea cu vîrfurile pătratului 2 triunghuri ascutitunghice și 2 Δ obtuzunghice



$$a^2 = b^2 + c^2 - 2bc \cos \hat{A} \Rightarrow b^2 + c^2 = a^2 + 2bc \cos \hat{A}$$

T. acutunghic

$$b^2 + c^2 < a^2 < 0$$

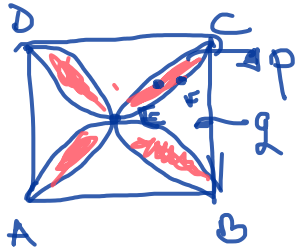


$$AF^2 + BF^2 < AB^2 \Leftrightarrow AF^2 + BF^2 < 1 \rightarrow \text{analog pt } \Delta BFC, DFC, ADF$$

$$nr_{\text{tri-obtuz}} = (AF^2 + BF^2 < 1) + (BF^2 + CF^2 < 1) + (DF^2 + CF^2 < 1) + (DF^2 + AF^2 < 1)$$

$$\text{cf } nr_{\text{tri-obtuz}} = 2.$$

OK



$$4p + 4q = 1$$

$$2p + q = \frac{\pi}{8}$$

(area unui semicerc) $1 \cdot 4$

$$\begin{cases} 4p + 4q = 1 \\ 8p + 4q = \frac{\pi}{2} \end{cases} \Rightarrow$$

$$4p = \frac{\pi}{2} - 1$$

$$4q = \frac{\pi}{2} - 1$$